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ABERT: Adapting BERT model for efficient detection of human and AI-generated fake news

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ABSTRACT

The proliferation of fake news in digital media poses a significant challenge to the dissemination of accurate information. Transfer learning, particularly with pre-trained language models (PLMs) like BERT, has demonstrated exceptional performance in natural language processing (NLP) tasks. However, the computational expense of fine-tuning the entire model for domain-specific tasks remains a limitation. In this study, we propose a novel approach, Adapt-BERT (ABERT), for the detection of both human and artificial intelligence (AI)-generated fake news. ABERT includes parameter-efficient adapter that enables efficient detection. By freezing the pre-trained BERT network and incorporating lightweight adapter, ABERT achieves comparable performance to fully fine-tuned BERT while reducing the number of trainable parameters by approximately 67.7%. ABERT strikes a balance between performance and computational efficiency, offering a scalable solution to combat the dissemination of fake news in digital media. Experimental evaluations on diverse datasets showcase the effectiveness of the proposed parameter-efficient approach in achieving comparable performance to state-of-the-art (SOTA) methods in the task of fake news detection (FND).

1. Introduction

The pervasive dissemination of fake news in today's digital landscape has become a pressing concern, posing significant challenges to the authenticity and reliability of information. Deliberately shared to mislead and cause harm, fake news has emerged as a critical issue. The volume of fake news has surged due to user-generated platforms and social media, which allow a wide spectrum of individuals, including those with malicious intentions, to rapidly create and spread misleading content. Fig. 1[1] illustrates the evolving patterns of interest surrounding the term 'fake news' over the past 12 months. Table 1 provides a summary of the key notations used in this paper for reference.

As the digital ecosystem evolves, it is important to recognize that the spread of fake news is not solely attributed to human actors but also involves the participation of AI systems. The authors in Tomassi, Falegnami, and Romano (2024) conducted a comprehensive analysis of information disorder in social media platforms, emphasizing the role of AI as both a spreader of fake news and an authoritative agent. Their study revealed that AI technologies, have significantly amplified the creation and dissemination of misinformation by automating many aspects of fake news generation. The interplay between human and AI-generated fake news has introduced new complexities and amplified the need for advanced detection methods. While human actors have long played a role in the propagation of misinformation, this phenomenon is not new; however, it has been amplified by the advent of digital technologies and social media platforms[2]. Malicious individuals intentionally create and disseminate deceptive content, exploiting the vulnerabilities of online social networks (OSNs). Their motives range from political agendas to economic gains, with far-reaching consequences, including sowing confusion and eroding public trust. The emergence of AI technologies has added a new dimension to the fake news problem. AI systems, powered by advanced algorithms and NLP capabilities, can generate highly convincing but entirely fabricated content. These AI-generated fake news articles, social media posts, and multimedia content are specifically designed to deceive and manipulate audiences. With the ability to produce vast volumes of misinformation at unprecedented speeds, AI systems have significantly amplified the challenge of combating fake news.

Assessing vast amounts of online data manually is highly challenging. To address this, researchers have made significant efforts to

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[1] https://trends.google.com/trends/.
[2] https://research.cbs.dk/en/publications/human-agency-in-the-propagation-of-false-information-a-conceptual.

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Fig. 1. Analyzing the trend of 'Fake News' interest over the past 12 months. The *x*-axis denotes key time intervals, and the *y*-axis represents relative search interest percentages.

Table 1
Summary of notations used in the paper.

Notation	Description
FND	Fake News Detection
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
PLMs	Pretrained Language Models
BERT	Bidirectional Encoder Representations from Transformers
SOTA	State-of-the-Art
OSNs	Online Social Networks
DL	Deep Learning
SVM	Support Vector Machine
CNNs	Convolutional Neural Networks
BiLSTM	Bidirectional Long Short-Term Memory
FFA	FeedForward Adapter
MHA	Multi-Head Attention

identify fake news by employing intelligent systems to analyze the content of news articles. Recently, machine learning (ML) techniques have been widely adopted for this purpose. In ML, detecting fake news typically involves utilizing either latent features, derived through neural networks, or non-latent features, which are hand-crafted and extracted from the content (Alghamdi, Luo, & Lin, 2024b). Researchers have dedicated significant effort to detecting fake news at the earliest possible stage by employing approaches that consider both the news content and its social context. While incorporating contextual information–such as user interactions on social media–may offer advantages, it also presents several challenges. These include the vast volume, noise, lack of structure, and incompleteness inherent in user-generated data. Furthermore, by the time such auxiliary information is collected, the fake news may have already been published and disseminated to a broad audience.

In this study, we focus on utilizing transfer learning as a methodology to extract valuable latent features, relying exclusively on the content of news articles for efficient and effective FND. Transfer learning is an ML technique in which a model trained on one task is adapted or fine-tuned for another, related task. This approach leverages the knowledge and features learned from the source task to enhance performance on the target task.

Transfer learning has demonstrated significant effectiveness in NLP, as it leverages pre-trained models to enhance performance across a wide range of tasks (Dai & Le, 2015; Devlin, Chang, Lee, & Toutanova, 2019; Howard & Ruder, 2018). Several studies have effectively utilized transfer learning in the context of FND, leveraging PLMs to improve the model's capacity to identify deceptive content (Alghamdi, Lin, & Luo, 2024a; Farokhian, Rafe, & Veisi, 2024). One common approach is to utilize feature-based transfer learning, where a PLM is employed as a feature extractor for capturing informative linguistic representations from news articles or social media posts. These representations are then fed into a downstream classifier or regression model, which is trained specifically for FND. By leveraging the knowledge acquired during pre-training, these models can effectively capture subtle linguistic cues and semantic patterns that are indicative of fake news. In contrast, PLMs can undergo a process of fine-tuning. This procedure entails taking a PLM and customizing it for the specific task of FND by training it on a domain-specific labeled dataset. This iterative fine-tuning process enables the model to refine its parameters and acquire task-specific features that are instrumental in distinguishing between authentic and deceptive news content. Fine-tuning offers the advantage of capturing domain-specific nuances and improving the model's performance on the target task. Furthermore, recent research suggests that fine-tuning these PLMs leads to even more superior outcomes compared to feature-based methods (Alghamdi, Lin, & Luo, 2023b). Both transfer learning techniques–namely, feature-based transfer and fine-tuning–possess distinct advantages and trade-offs. Feature-based transfer presents a lightweight solution that requires fewer computational resources, albeit potentially sacrificing the ability to capture task-specific nuances as effectively as fine-tuning. In contrast, fine-tuning offers the potential for higher performance but at the expense of increased computational requirements and a larger number of tuned parameters.

One notable example of these PLMs is BERT–a transformer network that stands for Bidirectional Encoder Representations from Transformers–which was trained on extensive text collections using unsupervised learning techniques. BERT achieved exceptional performance in tasks such as text classification and extractive question answering, surpassing previous benchmarks (Devlin et al., 2019). Recent studies have successfully employed BERT, either through feature-based methods or fine-tuning, to effectively detect fake news (Alghamdi, Lin, & Luo, 2022a, 2022b, 2023a; Jwa, Oh, Park, Kang, & Lim, 2019; Kaliyar, Goswami, & Narang, 2021; Liu, Wu et al., 2019), highlighting its strong performance in this domain. Although these studies have demonstrated SOTA results by leveraging the capabilities of BERT, it is worth noting that fully fine-tuning BERT can be computationally expensive, while feature-based approaches may not yield substantial performance improvements in comparison. To address the challenge of fine-tuning all model parameters while still achieving comparable performance across various NLP tasks, researchers have introduced several lightweight alternatives. These approaches focus on updating only a small subset of additional parameters, leaving the majority of the pre-trained parameters unchanged. For example, an adapter tuning approach was proposed in Houlsby et al. (2019), which introduces compact neural modules called adapters into each layer of the pre-trained network. During the fine-tuning process, only the adapters are trained.

Therefore, given the above considerations and the limitations associated with fully fine-tuning PLMs such as BERT, the central research question posed in this study is as follows:

- RQ: Is it possible to build an effective and efficient FND system that achieves performance comparable to fully fine-tuned BERT, while simultaneously reducing the number of trainable parameters?

To address the above question, the present study draws inspiration from the work of Houlsby et al. (2019) and builds upon existing efforts in the realm of FND. This work introduces ABERT, a novel approach and the primary contribution of the study. ABERT integrates adapter modules into BERT to tackle the task of FND. The central objective is to assess whether this approach can achieve performance comparable to that of fully fine-tuned BERT models, while substantially reducing the number of tunable parameters.

To this end, most current approaches are computationally intensive and require significant resources, which can be a limitation for many

academic researchers with only modest computational access. Furthermore, in real-world applications, datasets are likely to be much larger than the benchmark datasets typically used in research. To enable effective detection in such settings, a lightweight approach is essential. This study presents a lightweight yet highly effective adaptation of BERT. By leveraging parameter-efficient adapters, ABERT reduces the number of trainable parameters by approximately 67.7% while maintaining or surpassing the performance of fully fine-tuned BERT models. Extensive experiments validate its effectiveness, demonstrating accuracy and F1 scores that are comparable to or better than SOTA methods. Additionally, an ablation study highlights the contributions of individual components within the proposed model, further underscoring its utility. To the best of our knowledge, there is a scarcity of high-quality studies that explore parameter-efficient BERT adaptations specifically for FND.

2. Related work

2.1. Background

In recent years, OSNs have experienced a significant rise in popularity, emerging as accessible platforms for communication and information exchange. They have evolved beyond being mere channels for individual interaction and now play a crucial role in shaping public opinions and facilitating the dissemination of information. However, the proliferation of fake news on OSNs has become a pressing concern for both individuals and society at large. The absence of effective control, supervision, and automated fact-checking mechanisms has led to the generation of low-quality and fabricated content. Consequently, the general public is increasingly susceptible to disinformation and misinformation, including intentionally crafted fake news stories meant to mislead readers (Allcott & Gentzkow, 2017). Instances of false information spreading rapidly on OSNs, often with a degree of anonymity, have had tangible real-world impacts. Notable cases include a shooting incident triggered by fake news on Reddit and the prevalence of the "Pizzagate" fake news during the 2016 U.S. presidential election[3]. Research indicates that tweets containing falsified information on Twitter can reach users six times faster than trustworthy tweets, leading to fear, panic, and financial losses (Zubiaga, Aker, Bontcheva, Liakata, & Procter, 2018). The pervasive dissemination of fake news and its adverse social impact is a grave concern. Recognizing the urgency of addressing this issue, researchers in the field of NLP have devoted their efforts to developing ML and deep learning (DL) algorithms for the detection of fake news. In the upcoming section, we will introduce some related FND systems that have been proposed in the literature.

2.2. Methods for detection of fake news

Numerous approaches have been suggested in previous research studies regarding the detection of fake news. This section focuses on examining the methodologies that are most relevant to our work.

2.2.1. Statistical and advanced ML methods

Methods for identifying fake news typically center around analyzing the content of the news and the social context in which it is shared (Zhou & Zafarani, 2020). Content-based features primarily focus on extracting information from the text and visuals. (i) Textual features are those typically used in a traditional ML context. These features serve as the catalyst for describing content style with respect to four linguistic levels: lexicon, syntax, discourse, and semantic (Conroy, Rubin, & Chen, 2015). Existing studies have explored and extracted various linguistic-based features from the textual content of news articles for the purpose of detecting fake news. In a publication by Pérez-Rosas, Kleinberg, Lefevre, and Mihalcea (2017), the authors employed a set of linguistic-based features and conducted learning experiments to create robust fake news detectors. Their approach involved analyzing different aspects of the news articles, such as n-grams, LIWC (Linguistic Inquiry and Word Count), punctuation, grammar, and readability. Based on these features, a linear SVM (Support Vector Machine) classifier was trained. This research demonstrated the utility of computational linguistics in automatically identifying fake news. In another work by Fuller, Biros, and Wilson (2009), a linguistic-based method was proposed for detecting deception. The method involved utilizing 31 linguistic features, which were subsequently refined to only eight cues using three classifiers. Additionally, latent textual features (i.e., features utilized for representing news text) can be employed at different levels, such as the word level (Mikolov, Chen, Corrado, & Dean, 2013; Pennington, Socher, & Manning, 2014), sentence level (Arora, Liang, & Ma, 2017; Le & Mikolov, 2014), or document level (Le & Mikolov, 2014). These features generate vector representations that can be subsequently utilized for additional processing, such as serving as input for a classifier. In a research paper focused on identifying fake news through content analysis, Wang (2017) utilized convolutional neural networks (CNNs) and bidirectional long short-term memory (BiLSTM) to embed textual content and speaker metadata information for the purpose of detecting fake news. Similarly, Qian, Gong, Sharma, and Liu (2018) introduced a method involving two-level CNNs. In this approach, the first level of CNNs generates embeddings for individual sentences by considering the words, while the second level utilizes the sentence embeddings to produce an overall embedding for the entire article. (ii) Visual features, on the other hand, are derived from visual elements like images to capture distinct characteristics associated with fake news (Jin, Cao, Zhang, Zhou, & Tian, 2016).

In addition to analyzing content-based information, researchers have recognized the value of incorporating contextual features derived from users' social engagements on microblogging platforms to effectively identify fake news. These social engagements provide insights into the spread of news over time, offering additional information for determining the authenticity of news articles (Shu, Sliva, Wang, Tang, & Liu, 2017). Previous studies have predominantly focused on utilizing news content to model user behavior and interests, assuming that users' activities and posts on microblogs and social networks reflect their behavior. Context-based features encompass various aspects of users' interactions and engagements on social media platforms, including the number of friends (followers), posts, and replies (Zubiaga, Liakata, & Procter, 2016). Shu, Cui, Wang, Lee, and Liu (2019) proposed a framework that incorporates a sentence-comment co-attention subnetwork to jointly identify check-worthy sentences and comments in an explainable manner. However, it should be noted that their proposed framework, while effective, may not be readily adaptable to other real-world fake news datasets lacking user social comments, as the approach heavily relies on the availability of such data alongside news content.

Detecting fake news as early as possible is of paramount importance in today's information age. Misinformation can spread rapidly and have severe consequences, making it crucial to identify and combat it swiftly. However, relying on contextual user features for the detection process can lead to significant delays, as the collection of such features takes time. By the time these contextual user features are obtained, the fake news may have already reached a massive audience. The challenge lies in the limited availability of information about news propagation. Tracking the dissemination of news and monitoring its impact in real-time is a complex task. Obtaining user features, such as social media behavior or personal preferences, requires time-consuming data collection processes. Meanwhile, fake news can rapidly proliferate, potentially influencing and misleading a large number of people.

To address this issue, it can be beneficial to focus primarily on the news content itself, which is readily available. By analyzing the linguistic, contextualized patterns of news articles, automated systems can

[3] https://www.rollingstone.com/politics/politics-news/anatomy-of-a-fake-news-scandal-125877/.

be developed to detect fake news at an early stage. Content-based approaches leverage NLP techniques and DL models to identify suspicious patterns, misleading information, or deceptive tactics commonly used in the dissemination of fake news. Context-aware PLMs have gained recognition for their ability to capture deep contextualized information, leading to improved performance in FND. The following subsection introduces related research that employs contextualized models for this purpose.

2.2.2. Transfer learning using context-aware PLMs

Recently, the emergence of transformer-based architectures has propelled PLMs into the mainstream for text classification tasks (Devlin et al., 2019). Prominent advancements in this field have been driven by the utilization of PLMs such as ELMO (Peters et al., 2018), GPT (Radford, Narasimhan, Salimans, Sutskever, et al., 2018), and notably BERT and its variations, which have been trained on extensive corpora. Consequently, the question arises regarding how to leverage these models effectively for downstream text classification tasks. One approach involves incorporating task-specific layers onto the PLMs, which are then fine-tuned exclusively on those layers in a supervised manner (Devlin et al., 2019).

More specifically, these models employ a two-step learning process. Initially, they acquire language representations through pre-training, wherein a large amount of text is analyzed in a self-supervised manner. Subsequently, these pre-trained language representations can be applied to downstream NLP tasks using either a feature-based or fine-tuning approach. The former involves utilizing the pre-trained representations as additional features for learning the specific task, while the latter involves introducing minimal task-specific parameters and fine-tuning all pre-trained parameters on the downstream tasks. It has been shown that fine-tuning such PLMs is effective for many downstream tasks (Howard & Ruder, 2018). The key advantage of this transfer learning paradigm is the ability to acquire deep context-aware word representations from extensive unlabeled text corpora during self-supervised pre-training. This proves particularly valuable when dealing with domain-specific languages that have limited labeled data available (Alghamdi et al., 2023b).

Transfer learning approaches have revolutionized NLP tasks, including the detection of fake news. The researchers in Kaliyar et al. (2021) designed a DL architecture that incorporates the BERT model. BERT is used as a feature extractor, and additional layers are added on top to perform classification. The architecture included layers such as CNN, pooling layers, and fully connected layers. Kolluri and Murthy (2021) employed a system based on GPT-2 to evaluate news reliability and detect false information across diverse sources. However, the suitability of their system for processing large-scale datasets is limited by the considerable computational expense associated with GPT-2's extensive number of parameters. The Giant Language model Test Room (GLTR)[4], developed by Harvard University, is widely employed for the identification of machine-generated false information. Similarly, a study conducted by Zellers et al. (2020) explored the task of detecting machine-generated fake news. For more information on generative AI techniques, we refer to the study by Sakirin and Kusuma (2023), which provides further insights and resources on the topic. In their study, Alghamdi et al. (2022b) devised a sophisticated DL-based classifier, incorporating the BERT model, to categorize statements from the LIAR dataset into fine-grained categories of fake news. In Alghamdi et al. (2023a), the authors conducted a study focusing on the detection of fake news through the analysis of both news content and user posting behavior using the FakeNewsNet dataset. To capture deep contextual representations of news texts, the authors employed DL techniques, namely BERT, CNN, and a BiGRU model with a self-attention mechanism. These techniques were utilized to extract rich and meaningful features from the news content. However, a major challenge in applying transfer learning to FND is the demand for extensive computational resources (Strubell, Ganesh, & McCallum, 2019). A summary of the recent existing techniques/approaches used for FND is given in Table 2. These works highlight a gap in considering parameter efficiency, despite the growing importance of this aspect in designing effective and resource-conscious FND models.

While fine-tuning, a popular transfer learning technique, often outperforms feature-based transfer learning (Devlin et al., 2019), it can be computationally intensive. To address this challenge, recent research has focused on developing resource-lean approaches primarily based on the BERT architecture proposed by Devlin et al. (2019). These approaches aim to compress the BERT model while maintaining or even improving performance. A common strategy for model compression is knowledge distillation. Knowledge distillation, a resource-lean technique, involves transferring knowledge from a larger BERT model (the teacher model) to a smaller one (the student model). Sanh, Debut, Chaumond, and Wolf (2019) introduced a knowledge distillation method, DistilBERT, where the student model is trained to mimic the output probabilities and representations of the teacher model. This approach allows for a more compact student model while preserving the performance of the larger teacher model. Although these compressed models have proven effective in minimizing the size of architectural parameters, they are not without limitations. Their capabilities remain constrained by the underlying structure of the original BERT model, which limits their potential to achieve further parameter reduction (Liu, Rzepka, & Araki, 2021).

The process of adapting general-purpose PLMs for domain-specific tasks usually involves fine-tuning all model parameters, which can be excessively expensive, especially when dealing with large datasets. This becomes even more significant as PLMs continue to grow in size, ranging from hundreds of millions (Lewis et al., 2019; Radford et al., 2019) to hundreds of billions (Brown et al., 2020) or even trillions of parameters (Fedus, Zoph, & Shazeer, 2022). To address this challenge, several lightweight alternatives have been proposed, aiming to update only a small subset of additional parameters while keeping the majority of pre-trained parameters unchanged. One such approach is adapter tuning (Houlsby et al., 2019), which introduces compact neural modules called adapters into each layer of the pre-trained network. During the fine-tuning process, only the adapters are trained. Inspired by this approach, our proposed approach optimizes the allocation and utilization of parameters by injecting adapters into the BERT architecture, enabling greater flexibility in parameter reduction. This innovative approach aims to maximize parameter efficiency while achieving superior performance in FND.

The main objective of this study is to develop a parameter-efficient methodology that retains the high performance of fully fine-tuned transformer-based models while reducing computational demands. To evaluate the efficacy of the proposed approach, we focus specifically on the early-stage detection of fake news as it spreads on social media platforms. Our analysis is limited to the content of news articles in order to identify instances of fake news. This approach aligns with previous studies conducted by Galli, Masciari, Moscato, and Sperlí (2022), Gravanis, Vakali, Diamantaras, and Karadais (2019) and Silva, Santos, Almeida, and Pardo (2020), which also emphasize content-based analysis for FND. While social context features, such as user behavior and network relationships, have proven to be valuable in detecting fake news (Wang et al., 2020), our study does not specifically investigate or incorporate these features. Instead, our focus is on developing a parameter-efficient methodology for FND that emphasizes content analysis. For a comprehensive overview of recent approaches to FND, readers are referred to Alghamdi et al. (2024b).

In the following sections, we dive into the details of our proposed approach for FND, outlining the architecture, training process, and experimental results. We highlight the unique features and advantages of our approach and discuss its implications for improving the efficiency and effectiveness of FND systems.

[4] http://gltr.io/.

Table 2
Summary of existing techniques/approaches used for FND.

Ref.	Year	Dataset(s) used	Feature extraction	Context-aware?	Parameter-efficient
Ruchansky, Seo, and Liu (2017)	2017	Twitter and Webio	Doc2Vec	Yes	No
Singhania, Fernandez, and Rao (2017)	2017	PolitiFact	GloVe	No	No
Reis, Correia, Murai, Veloso, and Benevenuto (2019)	2019	BuzzFeed	BoWs	No	No
Shu et al. (2019)	2019	FakeNewsNet	GloVe	Yes	No
Gangireddy, Deepak, Long, and Chakraborty (2020)	2020	FakeNewsNet	Word2vec, Doc2Vec and Node2Vec	No	No
Dou, Shu, Xia, Yu, and Sun (2021)	2021	FakeNewsNet	Word2Vec and BERT	Yes	No
Shifath, Khan, and Islam (2021)	2021	Constraint@AAAI2021	Transformer-based models	Yes	No
Panda and Levitan (2021)	2021	Multilingual NLP4IF-2021 COVID19	BERT and BoW	Yes	No
Salem et al. (2021)	2021	FA-KES	GloVe and BERT	Yes	No
Raza (2021)	2021	NELA-GT-2020	BERT	Yes	No
Jarrahi and Safari (2021)	2021	FakeNewsNet	GloVe	No	No
Kaliyar et al. (2021)	2021	Twitter	GloVe and BERT	Yes	No
Li, Lu, Hu, Wang, and Lu (2022)	2022	FakeNewsNet	Word2Vec	No	No
Alghamdi et al. (2022b)	2022	LIAR	BERT	Yes	No
Raza and Ding (2022)	2022	NELA-GT-2019 and Fakeddit	BERT and GPT-2	Yes	No
Alghamdi et al. (2023b)	2023	Constraint@AAAI2021	Transformer-based models	Yes	No
Alghamdi et al. (2023a)	2023	FakeNewsNet	BERT	Yes	No
ABERT	–	**FakeNewsNet and AI data**	**BERT**	**Yes**	**Yes**

3. Research methodology

3.1. Problem statement

In the context of this research, a news article is classified as fake if it presents information that can be verified as incorrect. Given a labeled dataset of news articles, our goal is to develop an effective and efficient BERT model that can reduce the computational parameters required for full fine-tuning while capturing contextualized domain-specific features.

`Definition (FND)` The FND task is then defined as follows:

$$f : f(A) \rightarrow \{fake, real\} \ni, F(A) = \begin{cases} fake, & \text{if } A \text{ is fake content,} \\ real, & \text{otherwise.} \end{cases} \quad (1)$$

3.2. ABERT: Adapting BERT for efficient FND

By adapting BERT, we aim to leverage its pre-trained language representation capabilities while optimizing the model to achieve efficiency and reduce computational complexity. In this section, we dive into the implementation details and mechanisms of our proposed framework, ABERT. The overall architecture is illustrated in Fig. 2 (right). The FeedForward Adapter (FFA) (Section 3.2.2) is used for integrating the extracted features from the intermediate representations of the original BERT model and facilitating a comprehensive understanding of the input data. In the following subsections, we provide a detailed explanation of the implementation details of the architecture. We first provide an overview of the original BERT model, highlighting its key components and functionalities, as well as the objective of the adapter.

3.2.1. Original BERT structure and objective of auxiliary adapters

In this subsection, we explore the details of the original BERT architecture and outline the objectives of the auxiliary adapters. The core idea behind BERT is to learn contextualized word representations by considering both the left and right contexts of each word in a given sentence. This bidirectional approach allows BERT to capture the rich dependencies and relationships among words, resulting in more accurate and comprehensive language understanding. It is based on a multi-layer bidirectional transformer encoder architecture, incorporating an attention mechanism to capture contextual relationships between words or subwords within text. Unlike previous models that were primarily unidirectional, BERT leverages bidirectionality to better understand the meaning and context of words by considering both their preceding and succeeding words.

The encoding process in BERT involves several interconnected phases to transform the input text into meaningful representations. First, the input text is tokenized, breaking it down into individual tokens using WordPiece tokenization. Each token is then assigned an initial embedding vector, including special tokens like [CLS] and [SEP]. Positional encoding is applied to indicate the token order within the sequence. This encoding incorporates sinusoidal functions with different frequencies and amplitudes to capture positional information. The final token vectors then flow through encoder blocks, consisting of Multi-Head Attention (MHA) layers and feed-forward neural networks. The MHA layer calculates attention weights, while the feed-forward network applies non-linear transformations. Finally, BERT generates a vector representation of a certain hidden size for each token, capturing contextualized information based on its surrounding context. The [CLS] token's vector often serves as a summary representation for downstream tasks.

Mathematically, assume the input tokens are denoted as x, with shape (N, L), where N represents the batch size and L represents the maximum sequence length. The BERT model consists of multiple transformer blocks, where each block applies MHA followed by a feed-forward neural network. The output of the self-attention mechanism in the ith transformer block is denoted as H_i, with shape (N, L, D), where D represents the hidden size of the transformer. The MHA mechanism involves splitting H_i into multiple heads and performing separate linear projections for each head. Let us assume we have H attention heads. The attention heads are computed as follows:

$$H_i^h = W_{Qi}^h H_i, \quad H_i^{K^h} = W_{Ki}^h H_i, \quad H_i^{V^h} = W_{Vi}^h H_i, \quad (2)$$

where W_{Qi}^h, W_{Ki}^h, and W_{Vi}^h are the weight matrices associated with the query, key, and value projections for the hth attention head.

To compute the attention scores, we calculate the dot product between the query and key vectors, scaled by a factor of $\sqrt{D_h}$, and apply the softmax function:

$$A_i^h = \text{Softmax}\left(\frac{H_i^h (H_i^{K^h})^T}{\sqrt{D_h}}\right) \quad (3)$$

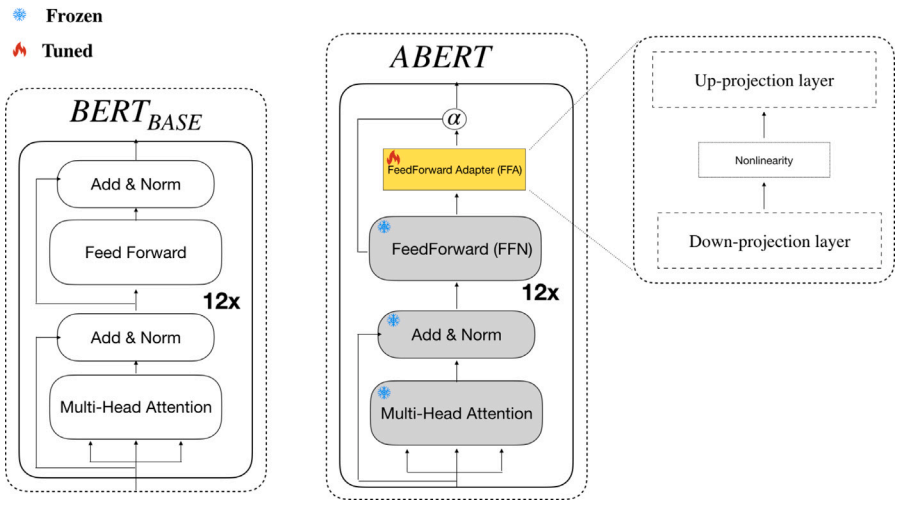

Fig. 2. High-level architecture of BERT (left) and ABERT (right). The adapter (yellow shaded block) consists of a bottleneck with a small number of parameters. During adapter tuning, the adapter is trained, including the final classification layer, while the BERT network is kept frozen.

where D_h represents the hidden size of each attention head. The attention scores A_i^h are then used to compute the weighted sum of the value vectors:

$$O_i^h = A_i^h H_i^{V^h} \tag{4}$$

Finally, the outputs of all attention heads are concatenated and linearly transformed to obtain the final output of the MHA layer:

$$O_i = \text{Linear}\left(\text{Concat}(O_i^1, O_i^2, \ldots, O_i^H)\right) \tag{5}$$

where Concat() represents the concatenation operation, and Linear() denotes a linear transformation.

Within BERT, the MHA mechanism enables the model to learn multiple attention patterns concurrently, referred to as heads. This parallel operation allows BERT to capture a wider spectrum of relationships between words. Since the attention heads have independent parameters, each head acquires a distinct attention pattern, contributing to the model's ability to comprehend diverse semantic associations among the words. Moreover, BERT utilizes the concept of pre-training and fine-tuning. During pre-training, BERT is trained on a large corpus using two unsupervised tasks: masked language modeling (MLM) and next sentence prediction (NSP). MLM involves randomly masking some words in the input sentence and training BERT to predict the masked words based on their context. NSP, on the other hand, involves determining whether two sentences appear consecutively or not. After pre-training, BERT can be fine-tuned on various downstream tasks, such as text classification, named entity recognition, and question answering. Fine-tuning involves training BERT on task-specific labeled data by adding task-specific layers on top of the pre-trained BERT model. The parameters of the entire model, including the pre-trained BERT layers and the task-specific layers, are updated during fine-tuning to adapt BERT's representations to the specific task in question. However, a drawback of large language models, such as BERT, is their requirement for fine-tuning on downstream tasks, which involves training the entire model with task-specific data. This process can be computationally expensive, particularly when dealing with large-scale datasets. To address this challenge, we propose the integration of auxiliary adapters within the BERT architecture.

The objective of the auxiliary adapters is to reduce the number of parameters in BERT while preserving its performance on downstream tasks. The adapters act as modular components that can be easily inserted into the BERT architecture, providing task-specific flexibility while leveraging the rich contextual representations learned during pre-training. The integration of these adapters offers an effective solution to reduce the computational requirements of fine-tuning while maintaining BERT's performance. Different layers of BERT capture various types of linguistic information. Research by Liu, Gardner, Belinkov, Peters and Smith (2019) demonstrated that different layers of the model capture distinct linguistic features. The earlier layers primarily focus on lower-level details, such as character-based features, while the middle layers tend to capture syntactic information, and the later layers emphasize semantic features (Jawahar, Sagot, & Seddah, 2019). Therefore, not all layers in a network contribute equally to improving the model's performance during training (Chen, Yuille, & Zhou, 2022; Zhang, Bengio, & Singer, 2022). The integration of adapters into different layers of BERT produces varying outcomes. This study found that incorporating adapters into all layers is essential, as discussed in Section 4.4.

In this study, our initial intention was to use the BERT$_{large}$ version as the base model. However, due to constraints imposed by limited computational resources, we decided to use the BERT$_{base}$ version as our base model. In the following subsection, we provide an in-depth explanation of the implementation details of our proposed model, including the structure of the FFA.

3.2.2. The proposed approach: ABERT

Our proposed model incorporates a highly effective module: the FFA with an adapter fusion approach. This enables the effective fusion of task-specific information with the representations learned by BERT, which is a key contribution of our work. Similar to the adapter module introduced by Houlsby et al. (2019), our model leverages an adapter to enhance BERT's performance on specific classification tasks, particularly FND in our case. The adapters reduce the original feature dimensions, which are in d dimensions, to a smaller, reduced dimension. After this reduction, a nonlinearity is applied, and the features are then projected back to their original d-dimensional space (Houlsby et al., 2019). By freezing the BERT encoder during the fine-tuning process and training only the adapter, our model achieves parameter efficiency while preserving the powerful capabilities of BERT. This approach allows us to optimize computational resources and streamline the training process for FND.

To incorporate nonlinearity into the adapter, we utilized the rectified linear unit (ReLU) as our chosen activation function (Li & Yuan, 2017). ReLU effectively eliminates negative values within an activation map by setting them to zero, thereby enhancing the expressiveness of the network. A notable advantage of ReLU lies in its non-saturating property, which significantly expedites the convergence of stochastic gradient descent compared to alternative activation functions (Li

& Yuan, 2017). Furthermore, ReLU demonstrates computational efficiency when compared to sigmoid or tanh activation functions, effectively addressing the challenge of vanishing gradients. The FFA is added after the feedforward layer in each block of the BERT encoder. It operates on the output of the feedforward layer, which contains the intermediate representations produced by the feedforward neural network within each block of the BERT encoder. The output of layer normalization is passed through an intermediate layer, followed by another layer, resulting in the final output.

$$S_i = FFN(O_i), \tag{6}$$

where O_i is the output of the MHA mechanism, FFN is the feedforward neural network that processes the output of the attention mechanism and S_i is the output of the feedforward neural network, representing the intermediate representations of the input sequence. The intermediate representation S_i is passed through the FFA, which contains two linear layers with a ReLU activation in between. This yields the output of the adapter F_i.

$$F_i = FFA(S_i), \tag{7}$$

The output of the FFA F_i is fused with the original intermediate representation S_i using a weighted sum, controlled by the learnable parameter α. By adjusting the value of α during training, the model learns to balance the importance of the adapter-based transformation and the original intermediate representation. Through this adaptive mechanism, the model effectively balances the relevance of the adapter-driven transformation with the inherent pre-trained knowledge within the BERT model. The fused representation G_i is calculated as follows:

$$G_i = \alpha \times F_i + (1 - \alpha) \times S_i \tag{8}$$

During training, the value of α is learned alongside the parameters of the adapter through backpropagation. This enables the model to find the optimal balance between the adapter's task-specific information and the pre-trained knowledge in the original BERT model. The fused representation G_i is then passed to the next layer (e.g., the next transformer block) for further processing. Our model processes the input through all transformer blocks, applying the FFA with the adapter fusion after each encoder block. This can be described as a learnable dynamic skip connection. While the static skip connection offers a straightforward information path, our adapter fusion presents a more flexible and adaptable mechanism: a dynamic skip connection. This dynamic approach, governed by a learnable parameter, allows the model to dynamically adjust the contribution of the original output and the adapter output during training. This makes the adapter's impact on the model more flexible and adaptable to different tasks or data distributions. If the adapter becomes more useful during training, the model can increase its influence (by learning a higher value for α), and if it becomes less useful, the model can reduce its influence. This dynamic behavior allows the model to learn when and how to utilize the adapter more effectively.

The purpose of FFA is to allow the model to learn additional task-specific information and optimize the intermediate representations specifically for the FND objective. By incorporating the adapter, the model can adapt the representations and capture relevant patterns and features for the detection task. The adapter's learnable parameters enable it to fine-tune the intermediate representations effectively. We aim to verify whether the adapter will deliver the same performance, so we use simple MLPs. We expect that using more advanced neural networks as the adapter will improve performance; however, they will have more trainable parameters, which contradicts the goal of maintaining a lightweight model. Conducting experiments to find the balance between the size of parameters and performance will be one direction for future work. In Section 4.2, a subsequent ablation study of ABERT is conducted for validation.

After all transformer blocks, the pooled output is taken from the BERT model, and a trainable classification head is added to perform binary classification.

$$Z = Classifier(Dropout(H_0[:, 0, :])), \tag{9}$$

where H_0 represents the output of the last transformer block. To compute the binary cross-entropy loss, we compare these logits with the true class label y. In this case, $y = 1$ represents the "real" class and $y = 0$ represents the "fake" class. Therefore, the binary cross-entropy loss can be calculated using the sigmoid function $\sigma(Z)$ applied to the logits as follows:

$$BinaryCrossEntropy(Z, y) = -ylog(\sigma(Z)) - (1 - y)log(1 - \sigma(Z)) \tag{10}$$

The $\sigma(Z)$ transforms the logits Z into a value between 0 and 1. The binary cross-entropy loss penalizes the model for incorrect predictions by measuring the dissimilarity between the predicted probability and the true label. The goal is to minimize this loss during training, improving the model's ability to correctly classify inputs as either "fake" or "real". The algorithm can be seen in 1.

3.3. Experimental setup

In this subsection, we provide an overview of the datasets used and baseline models. Then, we discuss the results of the proposed model compared to the baselines and the ablation study. The model's performance is evaluated using different evaluation metrics. We report the confusion matrix, which serves as a crucial evaluation tool for classification models on a known test dataset, providing valuable insights into their performance. It consists of four main elements: True Positives (TP), where positive cases are correctly classified as positive; False Positives (FP), representing negative cases incorrectly classified as positive; False Negatives (FN), indicating positive cases mistakenly classified as negative; and True Negatives (TN), where negative cases are correctly classified as negative. We also report the other common metrics: Accuracy (A%), Precision (P%), Recall (R%), and F1 (F1%) score.

3.3.1. Datasets

We use three real-world fake news datasets to perform parameter-efficient FND.

- FakeNewsNet[5]: is a widely recognized public dataset utilized for the detection of fake news. This dataset comprises news information from two reputable fact-checking websites, namely GossipCop[6] and PolitiFact.[7] The datasets encompass news content that has been annotated with labels by professional journalists, alongside additional contextual and temporal information. These two datasets cover different topics, allowing us to evaluate our model on a broad spectrum of topics. They are referred to as PolitiFact and GossipCop in this paper. The datasets can be accessed through the functionalities provided by the FakeNewsNet team and the Twitter API. The statistics of these datasets are shown in Table 3.
- AI-generated[8]: a dataset comprising approximately 387,000 text samples. These samples are sourced from various news articles available on the web, as well as collection of fake and real news generated by OpenAI's AI language model. It is referred to as the AI dataset in this paper. The dataset has been divided into three sets: train, validation, and test. Each set contains an equal distribution of the two classes, ensuring a balanced representation. For

[5] https://github.com/KaiDMML/FakeNewsNet.
[6] https://www.gossipcop.com/.
[7] https://www.politifact.com/.
[8] https://www.kaggle.com/datasets/bjoernjostein/fake-news-dataset?resource=downloa.

Algorithm 1 ABERT for FND.

Require: Input tokens x with shape (N, L), Number of attention heads H, Hidden size D, Training epoch $epoch$, Adapters and fusion parameters
Ensure: Predicted fake/legitimate probability $\hat{y}$
Input Representation:
$\boldsymbol{H_0} \leftarrow \text{Embed}(x)$ {Token embedding}
for $i = 0$ to $T - 1$ **do**
{For each transformer block i}
Multi-Head Attention (MHA):
$\boldsymbol{H}_{i,\text{MHA_input}} \leftarrow \boldsymbol{H}_i$ {Input to MHA for this block} {Details of MHA are as follows:}
for $h = 1$ to H **do**
$\boldsymbol{Q}_i^h \leftarrow W_{Qi}^h \cdot \boldsymbol{H}_{i,\text{MHA_input}}$ {Query projection}
$\boldsymbol{K}_i^h \leftarrow W_{Ki}^h \cdot \boldsymbol{H}_{i,\text{MHA_input}}$ {Key projection}
$\boldsymbol{V}_i^h \leftarrow W_{Vi}^h \cdot \boldsymbol{H}_{i,\text{MHA_input}}$ {Value projection}
Attention Scores:
$\boldsymbol{A}_i^h \leftarrow \text{Softmax}\left(\frac{\boldsymbol{Q}_i^h(\boldsymbol{K}_i^h)^T}{\sqrt{D_h}}\right)$ {Compute attention scores}
Attention Output:
$\boldsymbol{O}_i^h \leftarrow \boldsymbol{A}_i^h \cdot \boldsymbol{V}_i^h$ {Weighted sum of value vectors}
end for
Concatenate Heads:
$\boldsymbol{O}_i \leftarrow \text{Linear}(\text{Concat}(\boldsymbol{O}_i^1, \boldsymbol{O}_i^2, \ldots, \boldsymbol{O}_i^H))$ {Concat and linearly transform, output of MHA}
Feedforward Network (FFN):
$\boldsymbol{S}_i \leftarrow \text{FFN}(\boldsymbol{O}_i)$ {Output of the FFN after MHA}
Feedforward Adapter (FFA):
$\boldsymbol{F}_i \leftarrow \text{Linear}_2(\text{ReLU}(\text{Linear}_1(\boldsymbol{S}_i)))$ {Apply adapter with two linear layers and ReLU}
Adapter Fusion:
$\boldsymbol{G}_i \leftarrow \alpha \times \boldsymbol{F}_i + (1 - \alpha) \times \boldsymbol{S}_i$ {Fuse FFA output with original FFN output}
Layer Normalization:
$\boldsymbol{H}_{i+1} \leftarrow \text{LayerNorm}(\boldsymbol{G}_i)$ {Normalize for next block}
end for
Final Pooled Output:
$\boldsymbol{H_T} \leftarrow \boldsymbol{H_T}[:, 0, :]$ {Take pooled output (e.g., CLS token) from the last transformer block}
Classification Head:
$\hat{y} \leftarrow \sigma(\boldsymbol{W}_{cls} \cdot \boldsymbol{H_T} + b_{cls})$ {Fake/legitimate prediction from final pooled output}
Loss Computation (during training):
$\mathcal{L}_{cls} \leftarrow -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$ {Binary cross-entropy loss}
$\mathcal{L}_{total} \leftarrow \mathcal{L}_{cls}$ {Total loss (only classification loss)}
return $\hat{y}$

our experiments, we employed the train and validation datasets to train and evaluate our models, respectively. The statistics of the dataset are shown in Table 4.

3.3.2. Hyperparameters

We implement our model with PyTorch and conduct the training and evaluation using Google Colab on an NVIDIA Tesla T4 GPU. The FakeNewsNet dataset is utilized, with 80% of the data allocated for the training set and the remaining 20% for the test set. This division is commonly used in the literature. For the AI-generated dataset, we used the train and evaluation split provided by Kaggle. For our model architecture, we employed BERT's instance, `bert-base-uncased`. BERT consists of 12 transformer encoder layers with a dimensionality size of 867. The model is pretrained on a large corpus of text data and contains approximately 110 million parameters. During the training process, the model is optimized using the AdamW optimizer (Loshchilov & Hutter, 2017) with an initial learning rate of 1×10^{-5}, along with a weight decay of 10^{-2} (Loshchilov & Hutter, 2017). Additionally, a learning rate scheduler is utilized to track the accuracy during training and dynamically adjust the learning rate as needed. The hyperparameters were optimized through trial and error.

We observed that larger batch sizes led to memory issues; therefore, we decreased the batch size from the commonly used 32 to 8 to ensure smooth execution. We maintained this batch size of 8 throughout the training and validation process. After conducting experiments, we observed that training our proposed model for 20, 10, and 5 epochs for PolitiFact, GossipCop, and AI data, respectively, achieves optimal performance. On the other hand, the fully fine-tuned BERT model achieves its best performance when trained for 10 and 5 epochs for FakeNewsNet and AI data, respectively. Given the substantial size of our datasets, particularly GossipCop and the AI-generated dataset, the inherent presence of longer sequences necessitates our consideration of resource limitations. Consequently, we were unable to employ larger sequence lengths exceeding 128. Nevertheless, we anticipate that utilizing longer sequences would potentially yield improved performance, which we leave for future work. Therefore, in this work, we set the maximum sequence length to 128. Sequences longer than the maximum length were truncated, while shorter sequences were padded to reach the maximum length. The list specifying the different hyperparameters and their values is shown in Table 5.

Table 3
The statistical information of FakeNewsNet dataset.

Dataset	PolitiFact	GossipCop
No. Candidate News	694	18 676
No. True News	356	14 129
No. Fake News	338	4547

Table 4
The statistical data of the AI-generated dataset.

Dataset	Train	Validation
No. Candidate News	232 003	38 666
No. True News	164 751	27 458
No. Fake News	67 252	11 208

3.4. Evaluation metrics

For evaluating model performance, as mentioned previously, we employed four commonly utilized criteria in text classification tasks: Accuracy, Precision, Recall, and the F1 measure. These metrics are computed using the equations provided below.

$$Accuracy(A\%) = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

$$Precision(P\%) = \frac{TP}{TP + FP} \tag{12}$$

$$Recall(R\%) = \frac{TP}{TP + FN} \tag{13}$$

$$F1(F1\%) = \frac{2PrecisionRecall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN} \tag{14}$$

3.5. Baseline methods

- Text-CNN (Kim, 2014): Is a DL algorithm designed to facilitate text classification tasks. It leverages a combination of 1D convolutions and pooling layers to establish meaningful connections between the words within a given text, thereby capturing the underlying semantic relationships.
- HAN (Yang et al., 2016): Applies a hierarchical attention neural network framework to encode news contents, employing word-level attentions at the sentence level and sentence-level attentions

Table 5
Hyperparameters for ABERT.

Hyperparameter	Value
Maximum seq. length	128
Activation function	ReLU
Learning rate	1−e5
Optimizer	Adam
Loss function	Binary cross-entropy
Batch size	8
Number of epochs	20, 10 and 5

at the document level. This approach enables the model to effectively capture the intricate relationships and contextual information within the news content, allowing for enhanced analysis and understanding of the text.

- dEFEND (Shu et al., 2019): This framework proposed for detecting fake news consists of three primary constituents: a news content encoder, a user comment encoder, and a co-attention model that enables interaction between sentences and comments. The reported results are adopted from Shu et al. (2019).
- A Lite BERT (ALBERT) (Lan et al., 2020): Is a language representation model that improves upon BERT by enhancing model efficiency while maintaining or exceeding performance standards. It achieves this by utilizing parameter-sharing techniques, leading to a notable reduction in the number of model parameters and faster training and inference times. These characteristics make ALBERT highly suitable for environments with limited resources. Furthermore, ALBERT introduces cross-layer parameter sharing, enhancing the utilization of training data and overall model performance.
- eXtreme-Large Neural Network (XLNET) (Yang et al., 2020): Is an advanced NLP model that combines autoregressive language modeling and autoencoding, addressing their limitations. It surpasses models like BERT by maximizing the expected log likelihood across all possible permutations of factorization order.
- DistilBERT (Sanh, Debut, Chaumond, & Wolf, 2020): Is a highly efficient variant of the popular BERT model, which is roughly 40% smaller than BERT_{base}. It offers similar performance while significantly reducing the number of parameters.

The following models participate in the ablation study:

- BERT_{base}_x: It is a variant of the proposed framework where BERT is employed solely as a feature extractor, with its parameters frozen during training. It serves as a baseline for comparison.
- ABERT/FFA: It is a variant of the proposed framework that involves removing the effect of the adapter and fully fine-tuning the model (original BERT), meaning that all parameters of BERT are trained during the process. This model helps evaluate the impact of fine-tuning BERT on performance.

4. Analysis

4.1. FND performance

The experimental results presented in this section aim to validate the hypothesis stated in Section 1. We begin by applying our proposed model, ABERT, to various benchmark datasets, presented in Section 3.3.1, to assess its effectiveness in the task of FND. This evaluation serves as a confirmation of the model's performance and its ability to effectively and efficiently classify fake news instances. To further investigate the effectiveness of the individual components of our proposed model, we conduct an ablation study (Section 4.2). This study involves systematically removing or disabling specific components of the model and evaluating the impact on its performance. The performance of several models, including ABERT, ALBERT, text-CNN, HAN, dEFEND, XLNET, and DistilBERT, was evaluated on the PolitiFact, GossipCop, and AI datasets. Table 6 presents the weighted average results in terms of A(%), P(%), R(%), and F1(%).

Among all the models evaluated, ABERT achieves the highest weighted average performance on the PolitiFact dataset, outperforming all other models in terms of accuracy and precision. This indicates that ABERT is effective in accurately classifying fake news across multiple metrics on this particular dataset. On the GossipCop dataset, ABERT achieves performance comparable to DistilBERT, which has approximately 66 million trainable parameters, compared to ABERT's 39 million. Importantly, XLNet demonstrates the best performance on this dataset, reaching an accuracy rate of 89%. However, this comes at the cost of increased computational complexity due to XLNet's larger parameter size (117 million trainable parameters).

Although the reported performance of dEFEND cannot be directly compared due to differences in data acquisition time, experimental setup, and the type of input data used, it serves as a valuable reference for evaluating the proposed method. Despite not utilizing social user behavioral features–proven effective in prior studies (Alghamdi et al., 2023a)–our proposed method achieves comparable or even superior performance relative to SOTA methods. Specifically, our model demonstrates high accuracy on the FakeNewsNet dataset, surpassing the strong baseline dEFEND. This highlights the effectiveness of our approach, which achieves SOTA results by focusing solely on the textual content of news. We anticipate that incorporating additional context-based features, beyond the scope of this study, could further enhance our model's performance.

Our findings also indicate that the proposed model successfully captures intricate contextual information essential for distinguishing between fake and real news. On the GossipCop dataset, which relies exclusively on news content, ABERT demonstrates competitive results. However, it is worth noting that the performance of all models on this dataset remains below the 90% threshold. This underscores the potential importance of incorporating additional context-based information, such as user posting behavior, which we previously identified as effective in our earlier work (Alghamdi et al., 2023a), to achieve higher performance on this dataset.

On the AI dataset, ALBERT delivers promising results, while ABERT achieves performance comparable to DistilBERT and XLNET. This observation emphasizes that a model's effectiveness should not be solely judged by the number of trainable parameters. ABERT's ability to achieve competitive results, despite having fewer trainable parameters than both DistilBERT and XLNET, highlights the importance of considering other factors–beyond parameter count–when assessing model performance.

4.2. Ablation analysis of ABERT

We evaluated our model's overall performance across the classes using accuracy and weighted precision, recall, and F1 scores. All baseline models were trained using the same hyperparameters described in Table 5, unless otherwise stated. In the context of transfer learning, feature-based transfer uses a new function on top of pre-trained features, updating only the parameters of this new function. Fine-tuning, on the other hand, updates the parameters of the entire pre-trained model while introducing a task-specific output layer. Adapter tuning strikes a balance between these approaches by introducing additional parameters initialized to complement the original model and updating only these additional parameters during training. This method efficiently combines task-specific adaptation with maintaining the original pre-trained knowledge in a compact manner. The ablation study conducted in this subsection aims to evaluate the contribution of different components to the overall performance of the model. All datasets were evaluated using a consistent set of hyperparameters (see Section 3.3.2). The results of the ablation study, presented in Tables 7, 8, and 9,

Table 6
Performance comparison (%) of the baselines and proposed methods.

Datasets	Metric	ABERT	text-CNN	HAN	dEFEND	ALBERT	XLNET	DistilBERT
PolitiFact	A (%)	**0.9198**	0.7338	0.7338	0.904	0.9065	0.8921	0.8921
	P (%)	**0.920**	0.7364	0.7578	0.902	0.9151	0.8969	0.8935
	R (%)	0.9198	0.7338	0.7338	**0.956**	0.9065	0.8921	0.8921
	F1 (%)	0.9198	0.7258	0.7348	**0.928**	0.9070	0.8905	0.8924
GossipCop	A (%)	0.8579	0.8100	0.8231	0.80	0.8351	**0.8921**	0.8482
	P (%)	0.8582	0.7943	0.8145	0.729	0.8302	**0.8924**	0.8410
	R (%)	0.8579	0.8100	0.8231	0.782	0.8351	**0.8921**	0.8482
	F1 (%)	0.8434	0.7936	0.8170	0.755	0.8321	**0.8922**	0.8419
AI	A (%)	0.9909	0.9585	0.9519	NA	**0.9940**	0.9879	0.9762
	P (%)	0.9912	0.9597	0.9524	NA	**0.9941**	0.9884	0.9763
	R (%)	0.9909	0.9585	0.9519	NA	**0.9940**	0.9879	0.9762
	F1 (%)	0.9910	0.9577	0.9521	NA	**0.9940**	0.9880	0.9760

highlight the performance of different model configurations, with the best-performing models indicated in bold. These results provide valuable insights into the role of various components in enhancing the model's overall performance.

While each dataset possesses unique characteristics and properties, maintaining a consistent set of hyperparameters across all datasets ensures a fair and standardized evaluation. This approach isolates the effects of the model architecture, allowing variations in dataset properties to be accounted for during the analysis of results. To evaluate the effectiveness of different components, we conducted experiments with three variations of the proposed model, ABERT. The first variant, BERT$_{BASE}$_x, represents a frozen BERT model where all parameters remain unchanged during training, except for the classification head. As shown in Tables 7, 8, and 9, this variant struggled to capture complex relationships within the data, resulting in the lowest performance among the tested models. The second variant, ABERT/FFA, corresponds to a fully fine-tuned BERT model. While this variant demonstrated the ability to learn complex contextualized information from the data, it encountered challenges in identifying local salient patterns embedded within the data.

Upon evaluating these variations across different datasets, we observed the following results. On the PolitiFact dataset, BERT$_{BASE}$_x achieved an accuracy of 0.6750, while ABERT/FFA attained a higher accuracy of 0.9259, albeit at the cost of significant computational expense. In contrast, ABERT demonstrated a competitive accuracy of 0.9198, requiring a substantially lower number of trainable parameters. The confusion matrices for this dataset are shown in Fig. 3. Notably, our proposed model requires tuning only 39 million parameters to achieve competitive performance, which is significantly fewer than the full fine-tuned BERT model. Similar trends were observed on the GossipCop dataset. BERT$_{BASE}$_x achieved an accuracy of 0.7599, while ABERT/FFA reached 0.8522. The confusion matrices for this dataset are presented in Fig. 4.

On the FakeNewsNet dataset, ABERT, with approximately 39 million trainable parameters, performed on par with DistilBERT (see Table 6), which has around 66 million trainable parameters. This result underscores the effectiveness of our proposed model in achieving comparable performance to a condensed version of BERT while utilizing even fewer parameters. Similarly, on the AI dataset, BERT$_{BASE}$_x achieved an accuracy of 0.8038, while ABERT/FFA achieved an impressive accuracy of 0.9763. Remarkably, ABERT outperformed both models, achieving an accuracy of 0.9909. The confusion matrices for these results are shown in Fig. 5.

Through comprehensive evaluation and comparative experiments, we validate the effectiveness of our parameter-efficient model for FND. ABERT achieves comparable or superior performance to traditional transfer learning approaches while requiring significantly fewer computational resources. This not only addresses the challenge of expensive computational requirements but also provides a more efficient and scalable solution for FND in real-world applications. Furthermore, we found that the performance of both the proposed BERT and ABERT models is nearly identical, with ABERT requiring significantly fewer trainable parameters and progressively improving with more epochs.

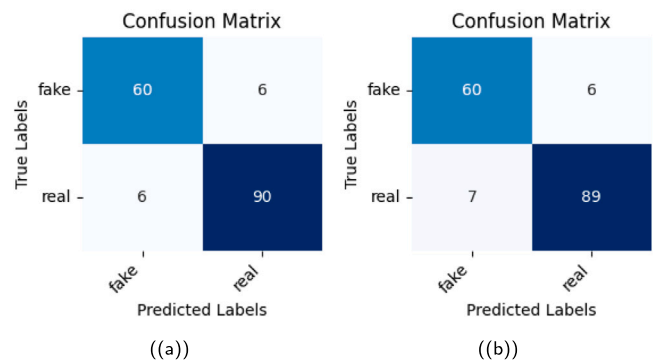

Fig. 3. (a) Full-Tuned BERT and (b) ABERT on PolitiFact dataset.

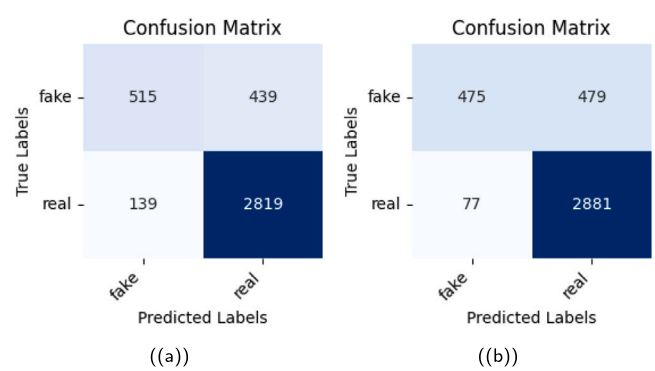

Fig. 4. (a) Full-Tuned BERT and (b) ABERT on GossipCop dataset.

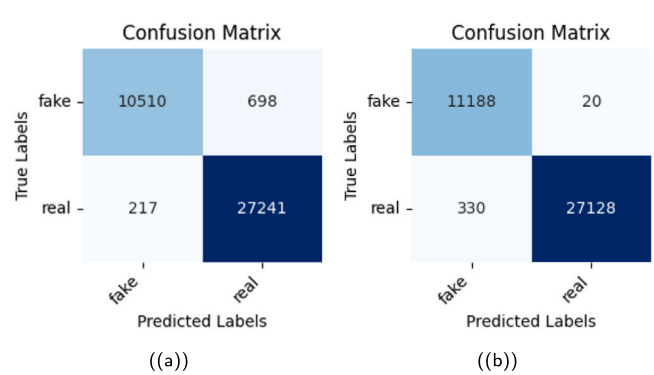

Fig. 5. (a) Full-Tuned BERT and (b) ABERT on AI generated dataset.

Table 7
The experimental results of the ablation study on PolitiFact.

Model	# Trainable param	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
$BERT_{BASE}$-x	1538	0.6605	0.6929	0.6605	0.6049
ABERT/FFA	109 483 778	0.9259	0.9259	0.9259	0.9259
ABERT	39 447 830	0.9198	0.9200	0.9198	0.9198

Table 8
The experimental results of the ablation study on GossipCop.

Model	# Trainable param	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
$BERT_{BASE}$-x	1538	0.7599	0.5775	0.7599	0.6562
ABERT/FFA	109 483 778	0.8522	0.8463	0.8522	0.8420
ABERT	39 447 830	0.8579	0.8582	0.8579	0.8434

Table 9
The experimental results of the ablation study on the AI dataset.

Model	# Trainable param	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
$BERT_{BASE}$-x	1538	0.8038	0.7983	0.8038	0.8000
ABERT/FFA	109 483 778	0.9763	0.9764	0.9763	0.9762
ABERT	39 447 830	0.9909	0.9912	0.9909	0.9910

Table 10
Comparison of model training and inference times, and loss for ABERT and ABERT/FFA on various datasets.

Dataset	Model	Training time (s)	Inference time (s)	Loss
PolitiFact	ABERT/FFA	40.05	9.29	0.2753
PolitiFact	ABERT	36.31	9.11	0.2971
GossipCop	ABERT/FFA	4199.72	311.71	0.3703
GossipCop	ABERT	3598.14	386.16	0.3813
AI data	ABERT/FFA	171 350.93	14 858.51	0.0751
AI data	ABERT	53 064.10	3693.65	0.0324

4.3. Training time and loss analysis

In this subsection, we provide a comprehensive analysis of the training and inference times, along with the corresponding loss values, for both the ABERT and ABERT/FFA models.

The findings (see Table 10) also reveal that the training (execution) time of the fully tuned BERT model on all datasets is longer than that of the proposed ABERT. In contrast, ABERT demonstrates shorter execution times overall; however, testing on the GossipCop dataset took slightly longer than with the fully tuned BERT model. We also report the test loss, with the overall results summarized in Table 10.

4.4. Assessing individual BERT layers with injected adapters

As previously mentioned, different layers of the model have the capacity to capture a wide range of linguistic information (Liu, Gardner et al., 2019). The lower layers primarily focus on finer details, such as character-based features, while the middle layers are more geared towards capturing syntactic information. In contrast, the higher layers emphasize the extraction of semantic features (Jawahar et al., 2019). In this subsection, we aim to explore alternative methods for improving performance while maintaining a reasonable model size. Specifically, we seek to assess the impact of different semantic contextualization levels within the ABERT model, based on various heuristics for utilizing BERT. This experiment aims to assess the impact of injecting the adapter into only k layers of BERT to evaluate how the model captures various levels of linguistic contextual information. The different variations tested are as follows:

- ABERT-FL1: In this variant, the adapter is injected into only the first lower layer of BERT.
- ABERT-LL1: Here, the adapter is injected into the last top layer of BERT.
- ABERT-FL4: In this variant, the adapter is added to the first lower four layers of BERT.
- ABERT-ML4: Here, the adapter is injected into the four middle layers of BERT.
- ABERT-LL4: In this variant, the adapter is added to the last top four layers of BERT.

Fig. 6 presents the performance analysis of the ABERT model when integrating the adapter into different numbers of layers (k). Notably, incorporating the adapter solely in the lower layers leads to a significant decline in performance on the PolitiFact dataset. However, the best performance is achieved when the adapter is integrated into the top k layers (where $k = 4$), indicating that these layers are crucial for capturing essential contextual information.

A similar trend is observed on the GossipCop dataset. When the adapter is placed only in the lower layers, there is a considerable performance drop. In contrast, placing the adapter in the middle k layers (where $k = 4$), which better capture semantic contextualized information, yields improved results. Fig. 7 shows the corresponding results on the GossipCop dataset, confirming this observation. These findings suggest that certain layers (especially those at lower levels in this analysis) may not contribute significantly to reducing the loss, and could potentially be skipped without adversely affecting overall performance.

Placing the adapter in the middle and top layers of the model yields better performance than using it in the lower layers. While the exact reason behind this phenomenon is not entirely clear, it is likely that higher layers capture more complex and contextually relevant information, whereas the lower layers primarily focus on basic features and initial representations. Similar trends are observed with the AI dataset (see Fig. 8), further supporting this hypothesis. By injecting the adapter at higher levels, the model can leverage the complex semantic context and fine-tuned information stored in those layers, resulting in enhanced performance.

4.4.1. Consecutive BERT layers

The selected layers exemplify distinct tiers of information extracted from the BERT model. The lower layers primarily encompass basic

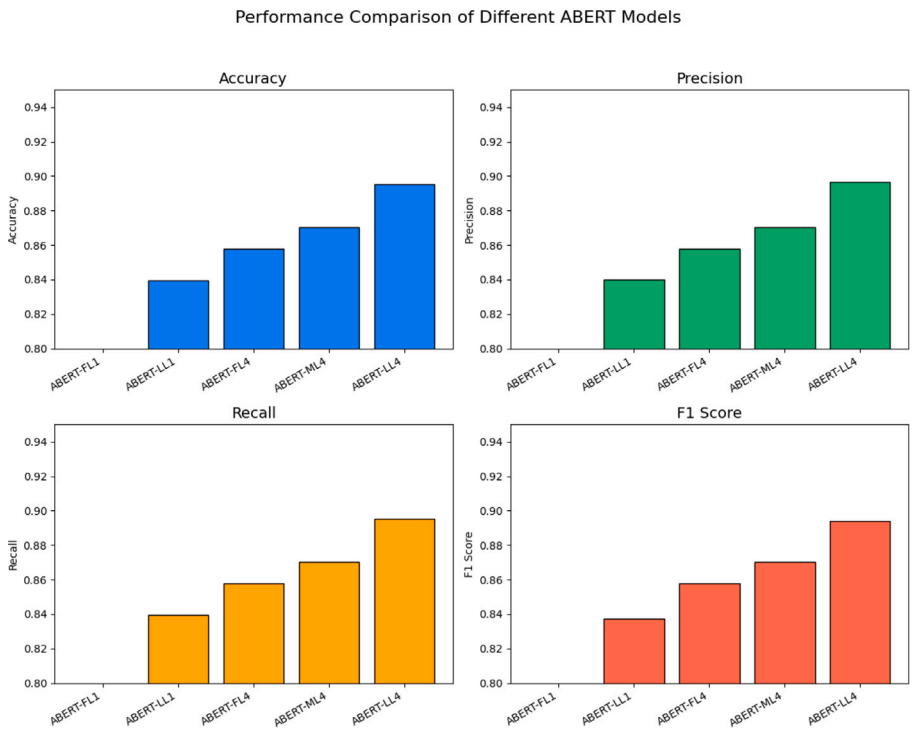

Fig. 6. Assessing individual BERT layers with injected adapters on PolitiFact.

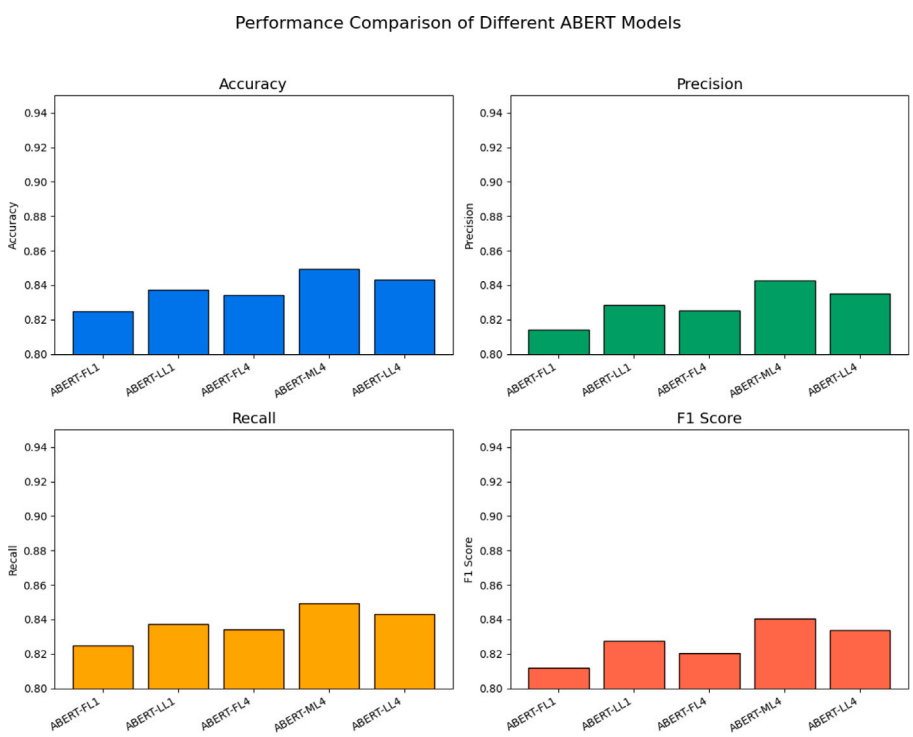

Fig. 7. Assessing individual BERT layers with injected adapters on GossipCop.

features and syntactic structures, while the intermediate and higher layers demonstrate a greater ability to capture intricate, contextually relevant information. By integrating adapters at these various levels, our goal was to investigate their adaptability across the different types of information embedded within the model. It is important to note that, in this study, we inject the adapter after a group of four layers, rather than after each individual layer, as we did in Section 4.4.

A study by Horne, Matti, Pourjafar, and Wang (2020) offers a significant insight regarding the arrangement of layers in the BERT model. Their findings highlight that contiguous clusters of four layers often exhibit similar informational attributes. This insight provides a compelling rationale for the improved performance observed when the adapter is added in our study. The strategic placement of adapters following groups of four layers allowed for specialized adaptation to the

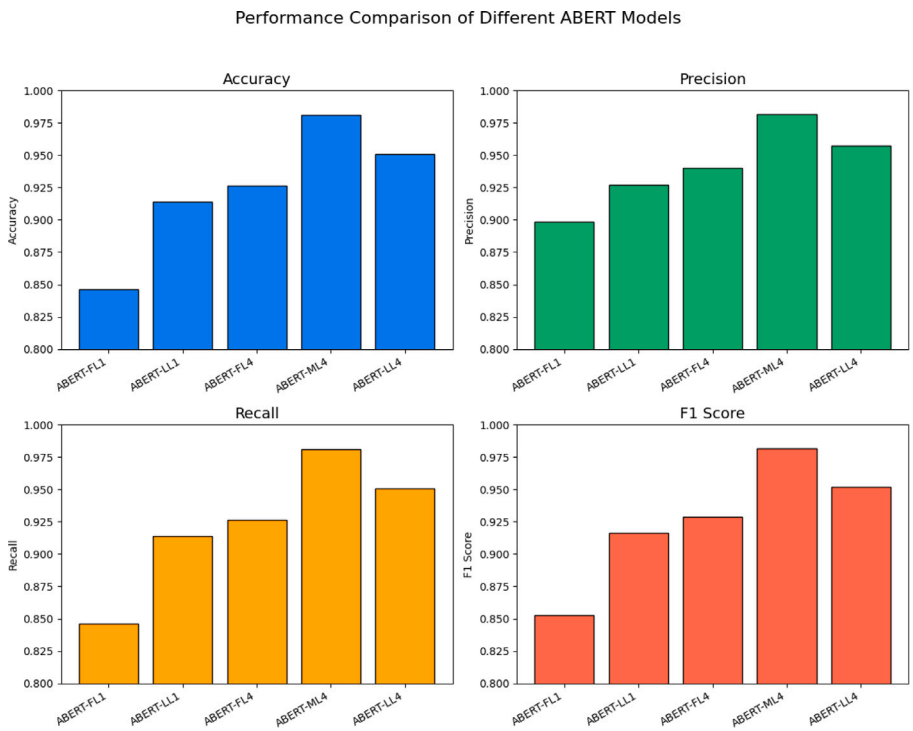

Fig. 8. Assessing individual BERT layers with injected adapters on AI dataset.

Table 11
Performance comparison (%) of different consecutive group of layers on PolitiFact.

Model	# Trainable param	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
ABERT	39 447 830	**0.9198**	**0.9200**	**0.9198**	**0.9198**
ABERT-con-2	25 273 622	0.8765	0.8798	0.8765	0.8745
ABERT-con-3	16 849 594	0.8704	0.8799	0.8704	0.8668
ABERT-con-4	12 637 580	0.8519	0.8519	0.8519	0.8519
ABERT-con-6	8 425 566	0.8642	0.8654	0.8642	0.8625

specific types of information captured by each group. More specifically, each set of four consecutive layers appears to exhibit a degree of coherence in the type of information they capture. By introducing adapters at these layers, we were able to focus on tailoring the model's representations to the distinct context and characteristics of that information. As a result, this targeted adaptation enabled the model to more effectively learn and leverage the patterns inherent in each layer's representation. This observation not only helps explain the behavior of the BERT model but also justifies our decision to use sets of four successive layers as the basis for adapter insertion.

In this subsection, we shift focus to a more detailed analysis aimed at evaluating the model's performance. This assessment, which involves integrating the FFA after various consecutive groups of layers, draws inspiration from the approach presented by Horne et al. (2020). We employ several variations of the ABERT model for this purpose:

- ABERT-con-2: This baseline involves injecting the adapter after two consecutive groups of layers in the BERT model.
- ABERT-con-3: In this variant, the adapter is integrated after three consecutive groups of layers.
- ABERT-con-4: Here, the adapter is inserted after four consecutive groups of layers.
- ABERT-con-6: This baseline adopts the adapter injection strategy after six consecutive groups of layers.

As shown in the tables, consistently superior results are achieved by injecting the adapter after each block–referred to as ABERT. In the context of the PolitiFact dataset, our investigation reveals that strategically placing the adapter after two or three consecutive groups of layers (ABERT-con-2 and ABERT-con-3, respectively) yields superior performance across key evaluation metrics, including accuracy, precision, recall, and F1 score. Notably, the ABERT-con-6 strategy, which involves adapter injection after six consecutive groups of layers, also demonstrates competitive performance while significantly reducing the number of trainable parameters. In contrast, performance declined when the adapter was injected after four consecutive groups of layers, indicating that this configuration may not be optimal for this dataset.

The efficacy of applying the adapter after two or three consecutive groups of layers can be attributed to its ability to capture more granular and task-specific information within the intermediate representations of the model. These intermediate representations encapsulate crucial context and semantics necessary for the downstream fact-checking task. Additionally, this injection strategy helps mitigate the risk of overfitting by introducing task-specific adjustments without disrupting the general representations learned by pre-trained BERT. The overall weighted average scores are provided in Table 11, with the best performance scores highlighted in bold.

For the GossipCop dataset, our findings indicate that the ABERT-con-4 strategy, which involves injecting the adapter after four consecutive groups of layers, achieves the second-best overall performance. This strategy allows the model to adapt effectively, capturing intricate patterns and nuances in the GossipCop data, resulting in higher scores compared to other strategies. While the ABERT-con-3 strategy, with adapter injection after three consecutive layers, also demonstrates performance improvements, it falls slightly short of the ABERT-con-4 approach. Conversely, the ABERT-con-6 strategy, despite its lower complexity and fewer trainable parameters, shows slightly reduced

Table 12
Performance comparison (%) of different consecutive group of layers on GossipCop.

Model	# Trainable param	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
ABERT	39 447 830	**0.8579**	**0.8582**	**0.8579**	**0.8434**
ABERT-con-2	25 273 622	0.8390	0.8330	0.8390	0.8231
ABERT-con-3	16 849 594	0.8428	0.8349	0.8428	0.8344
ABERT-con-4	12 637 580	0.8538	0.8494	0.8538	0.8419
ABERT-con-6	8 425 566	0.8280	0.8197	0.8280	0.8217

Table 13
Performance comparison (%) of different consecutive group of layers on AI data.

Model	# Trainable param	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
ABERT	39 447 830	**0.9909**	**0.9912**	**0.9909**	**0.9910**
ABERT-con-2	25 273 622	0.9685	0.9714	0.9685	0.9689
ABERT-con-3	16 849 594	0.9764	0.9779	0.9764	0.9767
ABERT-con-4	12 637 580	0.9535	0.9596	0.9535	0.9544
ABERT-con-6	8 425 566	0.9325	0.9450	0.9325	0.9343

performance compared to ABERT-con-4, suggesting that this strategy may not capture certain contextual details as effectively. The overall weighted average scores are presented in Table 12, with the best performance scores in bold.

In the AI dataset, we observed consistent performance improvements as the number of consecutive layers after which the MLP adapter is injected increases. This trend highlights the effectiveness of introducing more task-specific adaptations for AI-related content classification. The ABERT-con-3 strategy stands out as the best performer, achieving the highest accuracy, precision, recall, and F1 score while maintaining a balanced level of model complexity. Notably, the ABERT-con-2 and ABERT-con-4 strategies also yield similar performance, demonstrating the adaptability and flexibility of the adapter-based approach for AI data. The overall weighted average scores are provided in Table 13, with the best performance scores in bold.

Although class imbalance exists in datasets such as GossipCop and AI data, the proposed model successfully captures complex patterns and learns effectively despite these challenges. A key factor behind the improved performance is likely the model's ability to generalize and capture intricate text patterns. We hypothesize that by achieving a more balanced data distribution and incorporating context-based features, particularly for GossipCop data, the model's performance could be further enhanced. Despite class imbalance, the proposed model outperformed the fully fine-tuned BERT-based approach on both GossipCop and AI datasets, though the difference in performance was marginal. Moreover, the proposed model can achieve favorable results even when using small batch sizes, reducing the resource demands on hardware infrastructures.

To summarize, we have the following key findings:

- The ABERT architecture performed well on both small and large datasets, with only 39 million trainable parameters. It achieved performance comparable to fully fine-tuned BERT or the lightweight DistilBERT model, demonstrating its efficiency without sacrificing accuracy.
- Integrating the adapter into the model led to a significant reduction in classification error, highlighting the effectiveness of this approach in enhancing performance while maintaining a reduced parameter size.
- When the adapter was removed and BERT was used solely as a feature extractor, performance degraded. This indicates that the adapter's role is critical in fine-tuning the model to capture relevant task-specific features.

4.5. Error analysis

We opted for a qualitative error analysis, involving an assessment of the model's performance on test data to pinpoint patterns where it misclassifies labels (fake/real). Below, we present examples of errors from the test set as part of our evaluation. To gauge the limitations of our approach, we conducted an error analysis on classification results, revealing consistent misclassifications across different datasets. Table 14 displays instances within each dataset that our model inaccurately classifies.

- For the first error case, the model's inability to recognize the provided text as fake can be attributed to several factors. Firstly, the text's language and style closely resemble the structure and tone of legitimate news articles. If the model heavily relies on textual patterns associated with real news, it could struggle to differentiate this piece as fake. Moreover, the text comprises a blend of real-world entities, events, and claims, creating a challenge for the model to distinguish between fact and fiction. The inclusion of specific names, organizations (such as Wikileaks), and events (like the civil war in Syria) might create an illusion of authenticity. Furthermore, the text incorporates specific details like document and case numbers, along with explicit claims about the actions of prominent figures (such as Hillary Clinton and the Obama administration). These details could mislead the model into considering the content as legitimate due to the presence of apparent factual evidence.
- The model might have misclassified the fake story in the second case scenario as real due to the presence of some factual elements, such as mentioning President Trump's State of the Union address and references to real celebrities like Josh Gad and Jimmy Kimmel. Additionally, the mixture of real-world events and satirical elements in the text could have led the model to interpret it as a genuine news piece.
- In the third case, the misclassification could be attributed to the model's reliance on numerical data and statistics, which can sometimes lead to confusion when distinguishing between real and fake news. In this case, the presence of certain financial figures and industry-specific terminology may have skewed the model's decision towards classifying the text as real, despite its fabricated nature. It is worth noting that humans also struggle when confronted with similar situations. Notably, a class imbalance issue is present in the GossipCop and AI datasets, but addressing it falls beyond the scope of this study.

In the following section, we discuss the main findings of our research, highlighting their theoretical and practical implications. We also analyze the model's limitations and propose directions for future research.

5. Discussion

In this study, we aimed to develop an effective and efficient FND system that delivers performance comparable to a fully fine-tuned BERT

Table 14
A sample of misclassified examples obtained by ABERT model.

Dataset	Text	Actual	Pred
PolitiFact	"a leaked hillary clinton email confirms that the obama administration, with hillary at the helm, orchestrated a civil war in syria to benefit israel. the new wikileaks release shows the then secretary of state ordering a war in syria in order to overthrow the government and oust president assad, claiming it was the "best way to help israel". newobserveronline.com reports: the document was one of many unclassified by the us department of state under case number f-2014–20439, doc no. c05794498, following the uproar over clinton."	Fake	Real
GossipCop	"stars reacted with a mixture of anger, ridicule, skepticism and praise. president trump tried to strike an upbeat tone tuesday night for his first state of the union address, but he still drew strong reactions from celebrities on twitter. most stars reacted with a mixture of anger, ridicule, skepticism and sarcasm." this 2 h live broadcast of the new purge film on all of network tv is very bold, " josh gad mocked. late night comedians teed up their jokes early. jimmy kimmel watched the speech with stormy daniels, the adult film star who was reportedly paid to."	Fake	Real
AI data	"nextera energy partners lp : qtrly total operating revenues $176 million versus $161 million in the same period a year ago, compared with $116 million for the year — earlier period. qtrl total operating revenues $177 million vs. $160 million last year, primarily due to higher costs associated with energy and capital expenditures, and higher operating expenses related to the acquisition of nextera for $1. 3 billion in 2014."	Fake	Real

model while significantly reducing the number of trainable parameters. Through a series of experiments and subsequent analysis of the results, we draw the following conclusions.

5.1. Main findings

The results of our evaluation demonstrate that the proposed ABERT model achieves competitive performance across various datasets, matching or even surpassing fully fine-tuned BERT. ABERT delivers strong scores across all evaluation metrics, while significantly reducing the number of trainable parameters. On the PolitiFact dataset, ABERT achieves an accuracy of 0.9198, which is only 0.0061 (0.61%) lower than BERT's accuracy of 0.9259. This demonstrates that ABERT effectively classifies fake news on this dataset while using far fewer trainable parameters. Similarly, on the GossipCop dataset, ABERT outperforms both the fully fine-tuned BERT model and the lightweight DistilBERT, even though DistilBERT has considerably more trainable parameters. This highlights ABERT's ability to match the performance of more parameter-heavy models while maintaining efficiency and reducing computational complexity. On the AI dataset, ABERT again outperforms the fully fine-tuned BERT model and even surpasses DistilBERT, underscoring that model effectiveness should not be solely determined by the number of trainable parameters.

Furthermore, we conducted an ablation study to assess the importance of different injected layers (k) in improving performance. In this experiment, we applied the proposed adapter to specific layers while excluding its impact from others. The model's performance was then evaluated on the test set across various datasets. The results of this analysis are illustrated in Figs. 6, 7, and 8, showcasing the performance changes resulting from the integration and removal of the adapter from these layers.

Notably, we found that removing the adapter from the lower layers had only a minor impact on performance. However, when the adapter was removed from the middle and top layers, a significant decrease in performance was observed, especially on the PolitiFact dataset (e.g., FL1 variant). These outcomes align with the widely adopted fine-tuning strategy, where emphasis is often placed on the upper layers (Houlsby et al., 2019; Howard & Ruder, 2018) due to their ability to capture higher-level contextualized information. Our results support the common practice of focusing on adapting the upper layers to achieve better performance in fine-tuning scenarios. Additionally, the observation that removing the adapter from the lower layers had minimal impact is consistent with the understanding that lower layers typically capture more fundamental features, which are less task-specific and thus less affected by such adaptation. Interestingly, when the adapter's effect was removed from all layers of the network, the model's performance experienced a substantial drop (e.g., BERT_{BASE}_x).

These findings are in line with the study by Houlsby et al. (2019), which demonstrated that removing the adapters from lower layers has minimal impact on performance, while the absence of adapters from higher layers results in a significant performance decline. Based on these results, we conclude that the hypothesis raised in this work is valid. The findings indicate that it is indeed possible to build an effective and efficient FND system, such as ABERT, that achieves performance comparable to fully fine-tuned BERT while reducing the number of trainable parameters. ABERT thus presents a promising solution for the FND task, offering a balance of accuracy and efficiency.

The reduced number of trainable parameters in ABERT offers several advantages, including reduced computational requirements and faster inference times. Future research can explore techniques to further optimize and refine ABERT, as well as similar models, paving the way for more practical and efficient FND systems. Our proposed method has broad applicability, extending beyond the scope of FND to a wide range of NLP tasks. By leveraging and adapting pre-trained BERT models, our approach can be seamlessly applied to various NLP domains such as sentiment analysis, named entity recognition, text classification, and more. The flexibility and adaptability of our method allow researchers and practitioners to utilize the same framework and techniques to address different NLP challenges efficiently.

5.2. Theoretical implications

This study makes several significant theoretical contributions at the intersection of misinformation detection, PLMs, and parameter-efficient transfer learning. We begin by addressing a critical limitation in existing FND research—the high computational cost and inefficiency associated with fine-tuning large PLMs such as BERT and RoBERTa. While prior work (Alghamdi et al., 2022a, 2022b, 2023a; Jwa et al., 2019; Kaliyar et al., 2021; Liu, Wu et al., 2019) has established that these models perform well on various NLP tasks, their application in real-world, resource-constrained environments remains limited due to scalability concerns. Our proposed ABERT model directly extends this line of work by incorporating adapter layers, offering a parameter-efficient alternative that maintains performance while significantly reducing the number of trainable parameters. This approach builds

upon recent adapter-based innovations in NLP (Houlsby et al., 2019) and demonstrates their applicability in misinformation contexts, which has been relatively underexplored. Second, we extend the theoretical discourse on misinformation and FND by demonstrating how transfer learning can be tailored to differentiate between human-generated and AI-generated fake news—an emerging challenge in the digital misinformation landscape. Existing studies have focused primarily on traditional FND often overlooking the growing sophistication of AI-generated content. By including both content types in our experimental framework, we add to the theoretical understanding of how fake news manifests across different generation paradigms, contributing to a more nuanced conceptualization of FND in the post-GPT era. Third, our findings contribute to the literature on task-specific fine-tuning of PLMs by empirically validating that injection of adapter layers in the upper transformer blocks is particularly effective for FND. While prior studies have suggested this possibility (Horne et al., 2020), few have evaluated it systematically in misinformation-related tasks. By showing that this strategy not only reduces computational overhead but also achieves comparable or superior results to fully fine-tuned models, we extend both the architectural and theoretical understanding of efficient transfer learning. Finally, the novelty of ABERT lies in balancing computational efficiency and detection effectiveness, which offers a theoretical pathway for future research to consider both algorithmic accuracy and implementation feasibility.

5.3. Practical/managerial implications

The practical applications of the ABERT model are significant and extend across a variety of industries vulnerable to the propagation of fake news and misinformation. Misinformation can severely damage the reputations of organizations in sectors such as government, pharmaceuticals, etc., and even political parties, making the need for robust FND systems more critical. The ABERT model's efficiency in terms of computational power and memory makes it particularly well-suited for deployment in resource-constrained environments, where organizations may struggle with limited hardware infrastructure. For example, governmental organizations can use ABERT to combat the spread of fake news that undermines public trust, while pharmaceutical companies can apply it to protect the credibility of their products amidst misleading health claims. From a managerial perspective, this work highlights the importance of model efficiency and adaptability. By leveraging PLMs and incorporating lightweight adapters, businesses can implement robust FND systems that offer significant performance without the heavy computational cost typically associated with fine-tuning large models like BERT and RoBERTa. This enables organizations–regardless of their size or computational resources–to adopt scalable solutions for misinformation detection. Furthermore, these models can be adapted for other NLP tasks, expanding their applicability beyond FND to areas such as sentiment analysis, customer feedback analysis, and more.

5.4. Limitations

Although the proposed model is both effective and efficient, it still has some limitations. One key limitation is that we were unable to fully explain why injecting the adapters after certain layers in BERT led to better performance compared to other configurations. This challenge arises from the "black box" nature of the BERT model. In machine learning, a "black box" model refers to a complex algorithm or neural network whose internal workings are not easily interpretable or transparent. While we can observe the overall performance and outcomes of the model, understanding the exact decision-making process within the model is challenging. In our case, the adapter injection process involves intricate interactions between the adapter layers and BERT's layers, making it difficult to dissect the specific reasons behind the performance differences observed across configurations. The model's improvement in performance may be influenced by various factors, such as the flow of information, the contextualized features captured in specific layers, and how the adapter adapts and integrates this information.

In future work, we aim to develop a FND system that not only emphasizes high efficiency compared to a fully fine-tuned BERT model but also prioritizes interpretability. Additionally, while we have assessed the model on binary datasets so far, our future endeavors will involve evaluating its performance on multi-class and multilingual datasets.

6. Conclusion

This study introduced a parameter-efficient framework for FND, leveraging BERT to extract contextual features from textual data. Additionally, it demonstrated the practicality of the fusion adaptation technique. The proposed framework significantly reduces computational demands and decision-making time while maintaining favorable accuracy. The computational overhead associated with updating the ABERT model parameters is considerably reduced, resulting in more efficient and faster training compared to the BERT-based approach, which requires full parameter updates. Performance evaluation was conducted using accuracy, precision, recall, and F1 score on publicly available datasets, including FakeNewsNet (PolitiFact, GossipCop) and AI-generated data. The proposed framework achieved impressive results: 91.98% accuracy on PolitiFact, 85.79% on GossipCop, and 99.09% on AI, surpassing the performance of fully fine-tuned BERT. These outcomes underscore the effectiveness of the ABERT approach. While the presented framework outperforms current approaches and competes with fully fine-tuned BERT, the next step will involve incorporating social user context to enhance discrimination between true and false news. Furthermore, we aim to address the challenge of class imbalance in future work by exploring data augmentation techniques and innovative strategies for rebalancing class distributions. Additionally, we recognize the complexity of BERT-based models and plan to explore interpretability to improve model comprehension and accountability. We also intend to develop an efficient GPT model for FND. While GPT has a different architecture compared to BERT, our goal is to leverage GPT's capabilities in a way that optimizes both efficiency and performance in this context.

CRediT authorship contribution statement

Jawaher Alghamdi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuqing Lin:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Suhuai Luo:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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